Randomized trial of diet versus diet plus cardiovascular conditioning on glucose levels in gestational diabetes

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We studied the impact of a training program on glucose tolerance in gestational diabetes mellitus. Women with gestational diabetes mellitus (N = 19) were randomized into either group I, a 6-week diet alone group (24 to 30 kcal/kg/24 hours; 20% protein, 40% carbohydrate, 40% fat), or group II, which followed the same diet plus exercise (20 minutes three times a week for 6 weeks). An arm ergometer was used to maintain heart rate in the training range. Glycemic response was monitored by glycosylated hemoglobin, a 50 gm oral glucose challenge with a fasting and 1-hour plasma glucose, and blood glucose self-monitoring, fasting and 1 hour after meals. Week 1 glycemic parameters were the same for both groups. Week 6 data (mean ± SD) were as follows: group I glycosylated hemoglobin, 4.7% + 0.2% versus group II, 4.2% ± 0.2%; $p < 0.001$. The group I glucose challenge fasting value was 87.6 ± 6.2 versus 70.1 ± 6.6 mg/dl, $p < 0.001$ for group II. The group I 1-hour plasma glucose challenge result was 187.5 ± 12.9 mg/dl versus 105.9 ± 18.9 mg/dl for group II, $p < 0.001$. The glycemic levels diverged between the groups at week 4. We conclude that arm ergometer training is feasible in women with gestational diabetes mellitus and results in lower glycosylated hemoglobin, fasting, and 1-hour plasma glucose concentrations than diet alone. Arm ergometer training may provide a useful treatment option for women with gestational diabetes mellitus and may obviate insulin treatment. (Am J Obstet Gynecol 1989;161:415-9.)

Key words: Gestational diabetes, exercise, glucose tolerance, pregnancy, arm ergometry

Gestational diabetes mellitus is the most common medical complication of pregnancy. Current management consists of diet and careful monitoring of fasting and postprandial glucose levels. The goal of therapy is maintenance of euglycemia. When euglycemia is not achieved by diet alone, insulin therapy is recommended.[1] Gestational diabetes mellitus is considered in part to be a disease of glucose clearance, although it has been shown that this disorder is a heterogeneous entity.[2] Initiation of insulin therapy per se is palliative, but does not address the primary defect(s) that may include hyperinsulinemia. Rather, treatment modalities that overcome a peripheral resistance to insulin such as sulfonylurea treatment or exercise might be preferable. Although the sulfonylurea agents are reported to increase insulin sensitivity,[3] these agents are contraindicated for use in gestational diabetes in the United States.[4]

Cardiovascular conditioning exercise facilitates glucose utilization inter alia by increasing insulin binding to and affinity for its receptor.[5] Although exercise during pregnancy has been gaining acceptance,[6] it remains controversial.[7-15] Specifically, maternal exercise on a bicycle ergometer has been associated with fetal bradycardia.[7,11] Therefore we designed a study that used arm ergometry, which has not been associated with either uterine activity or fetal bradycardia,[16] to evaluate the impact of a training program on glucose tolerance in gestational diabetes mellitus.

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Material and methods

Subjects and study design. All women gave informed witnessed consent for the studies that were approved by appropriate institutional review. Table I characterizes the study population and infant outcome. Twenty untrained women with gestational diabetes diagnosed according to standard protocol[1] (Table I) were recruited. One woman was dropped from the study because she was found to have placenta previa and therefore could not be entered. Thus 19 women were randomized into two groups by drawing a number (group I or group II). The randomization process revealed that the peak plasma glucose response at 1 hour after a 50 gm glucose load was not significantly different in the two groups at the start of the program, whereas the peak 1-hour plasma glucose in group I on the 100 gm glucose tolerance test was significantly lower than in group II (Table I). The 50 gm glucose test with a fasting and 1-hour plasma glucose level will be referred to in this paper as the glucose challenge test.

Each woman was instructed to perform blood glucose

Table I. Clinical characteristics of nineteen study women and infant outcome

Patient No.	Glucose tolerance test at 28 wk (mg/dl)				Infant		Maternal age (yr)	C-peptide (pmol/ml)	Maternal weight at time of GTT (kg)
	Fasting	1 hr	2 hr	3 hr	Birth weight (gm)	Sex			
Group I									
1	90	199	168	115	3440	F	27	>3.00	67
2	85	207	169	133	3000	M	34	1.95	68
3	81	290	198	160	3100	F	32	2.75	93
4	90	191	168	152	4000	M	36	>3.00	82
5	107	223	185	162	3760	M	28	>3.00	70
6	98	217	168	148	3520	F	29	1.87	73
7	86	192	184	168	3370	M	31	2.43	79
8	85	201	190	145	3820	F	31	2.75	74
9	94	272	201	162	3170	F	32	>3.00	72
		Mean 221.3			Mean 3465		Mean 31.1		Mean 75.3
		SD 35.7			SD 343		SD 2.8		SD 8.2
Group II									
1	93	275	201	182	3560	F	27	>3.00	64
2	95	293	240	128	3212	M	26	>3.00	98
3	87	299	270	165	3473	F	29	1.95	82
4	106	282	180	120	3870	F	29	2.71	84
5	102	215	169	148	4000	M	31	2.53	87
6	87	275	170	122	3980	F	30	>3.00	72
7	84	263	172	117	3900	M	27	2.10	71
8	97	259	164	148	3127	M	30	1.98	73
8	98	245	187	136	3771	F	34	>3.00	79
10	102	229	192	108	3443	M	32	>3.00	74
		Mean 263.5			Mean 3634		Mean 29.5		Mean 78.4
		SD 27.2			SD 317		SD 2.5		SD 9.7
		$p = 0.01$*			$p = 0.282$*		$p = 0.203$*		$p = 0.471$*

GTT, glucose tolerance test.
*Group I versus group II.

self-monitoring four times daily (before breakfast and 1 hour after each meal). Each week all subjects were seen by a physician. Insulin was to be initiated if the fasting plasma glucose level rose higher than 105 mg/dl[1] and/or the 1-hour postprandial plasma glucose level was >140 mg/dl.[17] Serial evaluation of the glucose challenge test included a fasting plasma glucose, a 50 gm oral glucose load, and a 1-hour plasma glucose study. Plasma glucose was obtained by venipuncture and analyzed by the glucose oxidase method. The glucose challenge test was performed before and after the 6-week protocol. A fasting C-peptide specimen was drawn with the fasting plasma glucose determinations.

Group I was managed for 6 weeks with a standard diet that consisted of 40% carbohydrate, 20% protein, and 40% fat calculated at 24 to 30 kcal/kg/24 hours present pregnant weight divided into three meals and three snacks.[18] These women did not participate in any structured exercise program. Weekly questionnaires revealed that their only form of exercise was cleaning, shopping, and "running after toddlers." Group II received the same diet and participated in an exercise program of 6 weeks' duration that consisted of 20 minutes of supervised aerobic training on an arm ergometer (UBE-Cybex Fitness, Ronkonkoma, N.Y.). Six women exercised in the afternoon, between 4 PM and 5 PM (3 hours after the last meal) and four of the women exercised in the morning between 10 and 11 AM (3 hours after breakfast). These times were chosen because the blood glucose levels in the diet alone group were stable and comparable at these time points.

The patients were monitored for 20 minutes of cardiovascular work during which they maintained their target heart rate,[11] which was calculated to be equal to (220 − age in years) × 70% unless this heart rate exceeded 140 beats/min; then 140 beats/min was the target heart rate.[6] As each patient adapted to her workload in terms of perceived exertion and heart rate, the workload (or resistance) (kg × meters × min^{-1}) was increased or decreased by 10% to 20%; thus maternal heart rate was maintained in the training range. The exercise session never exceeded 50% maximal oxygen consumption ($\dot{V}O_{2\,max}$) in any case. Fifty percent oxygen consumption was estimated based on our previous work that related maternal heart rate in beats/min to the $\dot{V}O_2$ at steady state utilizing a metabolic cart (Beckman model S/N 614, D/N 11265, Fullerton, Calif.).[11] Blood glucose levels were measured before and after each exercise session. In over 150 sessions of exercise with the arm ergometer with pregnant women, no fetal

Table II. Group I data before and after 6-week dietary program

Patient No.	Plasma glucose before diet		Plasma glucose 6 wk after diet	
	Fasting	1 hr after 50 gm oral glucose (mg/dl)	Fasting	1 hr after 50 gm oral glucose (mg/dl)
1	92	194	90	174
2	105	195	94	179
3	110	215	98	168
4	117	257	82	184
5	105	285	80	190
6	98	247	92	210
7	82	205	82	192
8	79	240	87	192
9	92	195	84	199
Mean	97.8	225.8	87.6	187.5
SD	12.7	32.8	6.2	12.9
	$p = 0.048$		$p = 0.005$	

Table III. Group II data before and after 6-week dietary program

Patient No.	Plasma glucose before diet		Plasma glucose after 6 wk of training	
	Fasting	1 hr after 50 gm oral glucose (mg/dl)	Fasting	1 hr after 50 gm oral glucose (mg/dl)
1	90	240	60	88
2	95	270	70	77
3	110	195	64	105
4	120	210	77	103
5	120	220	82	142
6	95	240	74	118
7	102	197	69	88
8	98	276	73	110
9	90	247	65	123
10	104	213	67	105
Mean	100.4	230.8	70.1	105.9
SD	9.3	28.5	6.6	18.9
	$p = 0.000$		$p = 0.000$	

bradycardia or uterine activity has been observed. The caloric expenditure associated with this form of exercise is equal to the expenditure on a bicycle ergometer at the same heart rate for the same duration, provided that the test method utilizes the same muscle groups trained by the regimen.[19]

Laboratory evaluation. The plasma glucose level was measured by glucose oxidase (Yellow Springs Instruments, Yellow Springs, Ohio). Glycosylated hemoglobin was analyzed by high-pressure liquid chromatography.[20] The normal range for glycosylated hemoglobin in pregnancy is 3.5% to 4.5%, and the mean glycosylated hemoglobin is 4.0 ± 0.250, based on 100 pregnant women who had normal glucose tolerance tests. C-peptide was analyzed by radioimmunoassay according to the manufacturer's instructions (Novo Biolabs Antiserum M1221, Bagsverd, Denmark).[21] The normal range in pregnancy is 0.5 to 2.0 pmol/ml.

Fetal heart rate and uterine activity were measured continuously for 10 minutes before, during, and after exercise with a fetal monitor (Corometrics model 111 fetal monitor, Wallingford, Conn.). Uterine activity was expressed as a percent of total exercise time (minutes of contraction/total minutes of exercise × 100). A contraction was defined as a tocometer deflection >15 mm Hg above baseline.

Statistical analysis. Statistical analysis used the STATWORKS program on the Macintosh PC. Paired or unpaired t tests were used to analyze the data as appropriate. All mean values are expressed as ± 1 SD.

Results

Groups I and II did not differ (Table I) in maternal age (31.1 ± 2.8 vs. 29.5 ± 2.5 years, $p = 0.203$), maternal weight at the start of the program (75.3 ± 8.2 vs. 78.4 ± 9.7 kg, $p = 0.471$), maternal C-peptide level (range 1.87 to 3.0 vs. 1.95 to 3.0 pmol/ml), gestational age at delivery (range 39.4 to 40.0 vs. 39.5 to 40.5 weeks), neonatal birth weight (3465 ± 343 vs. 3634 ± 317 gm, $p = 0.282$), or gender of the infants (5 females

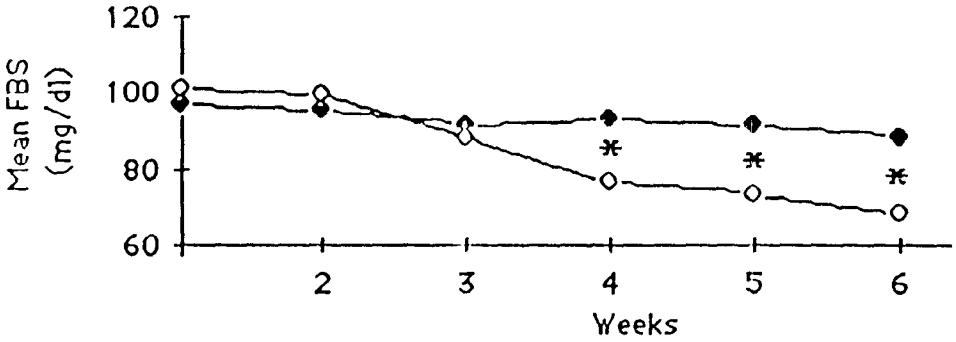

Fig. 1. Weekly mean fasting blood glucose (milligrams per deciliter) of blood glucose self-monitored results from week 1 to week 6 of protocol. *Closed circles* (●) represent mean values for group I (diet alone) and *open circles* (○) represent group II (diet and exercise). *Asterisk* signifies significance of $p < 0.001$.

vs. 5 females). The 1-hour plasma glucose value on the diagnostic glucose tolerance test did differ between the groups: group I, 221.3 ± 35.7 vs. group II, 263.5 ± 27.2 mg/dl ($p = 0.01$).

Despite the difference in the diagnostic glucose tolerance test results, at the beginning of the protocol groups I and II had equivalent 1-hour blood glucose responses (Tables II and III) to a glucose challenge test (group I = 225.8 ± 32.8 mg/dl vs. group II = 230.8 ± 28.5 mg/dl, p = NS). After 6 weeks, the mean 1-hour glucose response in group I on the glucose challenge test was 187.5 ± 12.9 mg/dl compared with the women in group II, who had a mean 1-hour response of 105.9 ± 18.9 mg/dl ($p < 0.001$, group I vs. group II). Group I did improve their 1-hour glucose response to a challenge with the program of diet alone (initial vs. final $p = 0.005$) but the mean 1-hour glucose response on the glucose challenge test was still greater than the current cutoff for the initial screening value for the diagnosis of gestational diabetes of 140 mg/dl[1] (Table II).

The chronic effect of exercise on hepatic glucose output was reflected by the fasting plasma glucose levels on the glucose challenge tests before and after the 6-week protocol. The mean fasting plasma glucose value in group I was 97.8 ± 12.7 mg/dl during the first glucose challenge test and 87.6 ± 6.2 mg/dl on the glucose challenge test after 6 weeks of diet alone ($p = 0.048$). The mean initial fasting plasma glucose value on the glucose challenge test in group II was 100.4 ± 9.3 mg/dl ($p = 0.478$ group I vs. group II). After 18 training sessions, the mean final fasting plasma glucose on the glucose challenge test performed between 15 to 45 hours after the eighteenth session for group II was 70.1 ± 6.6 mg/dl ($p < 0.001$ initial fasting plasma glucose in group II vs. the final). The mean fasting plasma glucose on the final glucose challenge test in group II was also significantly lower than the final value for group I ($p < 0.001$).

The glycosylated hemoglobin level improved in both groups. In group I the mean initial glycosylated hemoglobin value was 4.86% ± 0.39%; after 6 weeks this level fell to 4.67% ± 0.25% ($p = 0.01$). In group II the initial glycosylated hemoglobin value was 4.83% ± 0.33%, which fell to 4.16% ± 0.28% ($p < 0.001$). The glycosylated hemoglobin values for the two groups were not different at the start ($p = 0.75$) but there was a significant difference ($p < 0.001$) after 6 weeks.

To determine the time course of the glucose changes in group I and group II, the daily blood glucose self-monitoring of fasting values for each group were evaluated (Fig. 1). By 4 weeks of the protocol, the mean fasting blood glucose for group II (75.8 ± 5.8 mg/dl) diverged significantly ($p < 0.001$) from that of group I (93.1 ± 6.5 mg/dl). The postprandial glucose levels on self blood glucose monitors in group I were higher than those of group II at the 6-week point (136 ± 10 mg/dl vs. 112 ± 10 mg/dl, $p < 0.01$).

The 10 women who trained (group II) showed a continuing and significant change in glucose response to exercise. On the first day of the training program, the mean glucose decrement during exercise expressed as a percent was 8.4% ± 3.0%. During the eighteenth and final training session, the mean decrement was 29.8% ± 8.0% ($p < 0.001$, initial vs. final percent decrease associated with a training session).

There were no episodes of fetal bradycardia or uterine activity noted during any exercise session. No episodes of hypoglycemia (blood glucose <60 mg/dl) occurred in the study population. None of the 19 women in group I or group II required insulin. Of note, the woman with gestational diabetes who was dropped from the study and who was placed on strict bed rest as a result of placenta previa, did require insulin. All 19 infants born to this study were healthy, term neonates with no signs of morbidity.

Comment

Our study documents that women with gestational diabetes can train with arm ergometry and that such a

program of cardiovascular conditioning exercise results in lower levels of glycemia than a program of diet alone. The effects of exercise on glucose metabolism became apparent after 4 weeks of training and appeared to impact hepatic glucose output (as reflected by fasting glucose levels), glucose clearance (as reflected by glucose values after a 50-gm oral glucose challenge), and mean blood glucose (as reflected by glycosylated hemoglobin values). These combined observations would be consistent with a long-term effect of exercise rather than a short-term effect from individual sessions. The important question of the duration of a training effect on glucose metabolism in women with gestational diabetes was not addressed in this study.

Our observations appear contrary to the results in the literature asserting that the natural progression of glucose intolerance during gestation is deterioration with increasing gestational week[22] and that walking exercise in type I insulin-dependent diabetic pregnant women is associated with no improvement.[23] However, neither of the above studies addressed the issue of gestational diabetes with its unique problems of glucose homeostasis.

The implications of our study for the health care of women with gestational diabetes remain to be determined. Compliance with the protocol was 100% and there was no maternal or infant morbidity associated with the training program in this small series. Our results also imply that a cardiovascular conditioning program might obviate insulin treatment in many women with gestational diabetes. Therefore the economic and health care implications of these observations warrant further testing.

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